

solid: construction steps

dashed: the two certificates

1 extract

features and labels from disjoint streams; assert that no feature names the label

2 diagnostic

fit a position-only model on the whole corpus

3 cases

windows of W units ending Lead units before an event

4 controls

k per case from the same recording unit, balanced before and after

5 matched sets

disjoint sets of a case and its controls

6a certificate: P3

position-only AUC on the matched sample; grant if its 95% CI includes 0.5

6b certificate: P4

$D(L) = \text{AUC}(L-1) - \text{AUC}(L)$; certify at the smallest L with $D(L+1)$ below the threshold

7 evaluate

grouped leave-one-participant-out; pooled AUC, participant-clustered CI

8 permutation null

permute case status within matched sets

9 sensitivity

repeat over a grid of W, g, m, k and seed

10 report

pooled and fold-wise results, paired tests, grid range